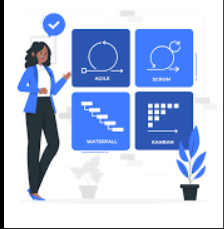


COMPLETE SURVEY ON SOFTWARE ARCHITECTURAL STYLES AND PATTERNS

Elaborado por: Ariadna Cardona, Julián Quiroz y Samuel Tarazona
Arquitectura de Software.
Ingeniería de Software
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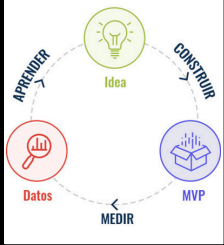
ARCHITECTURE AS FOUNDATION

Software architecture provides the structural blueprint for meeting system requirements across its lifecycle.



STYLES VS. PATTERNS

Architectural styles define system structure (e.g., layered, client-server), while patterns offer reusable solutions to recurring design problems.



QUALITY-BASED COMPARISON

Styles are compared by attributes like performance, scalability, maintainability, and security.



MODERN ARCHITECTURES COVERED

- Includes recent trends like microservices and cloud-based architectures.



COMPREHENSIVE LEARNING RESOURCE

Serves as a useful reference for both practitioners and researchers in software engineering.



BALANCE STYLE AND TRADE-OFFS

Design decisions must carefully balance adherence to architectural styles with competing goals such as flexibility, scalability, or resource efficiency—highlighting the importance of trade-off analysis.

